PFA Internship Report

2nd Year Computer Engineering

Automating the IT Service Request Lifecycle

(Request Fulfillment) Using ServiceNow

Internship Period

2 July 2026 – 31 August 2026

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Acknowledgments

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I would especially like to thank Houda AJDIR, my company supervisor, for her guidance, availability, and continuous support throughout the internship. Despite the remote nature of the internship, her explanations, feedback, and technical guidance were essential to understanding the project requirements, addressing the challenges encountered, and progressing effectively throughout the different stages of the work.

The remote nature of the internship also encouraged me to work independently, organize my tasks effectively, manage my time, and take responsibility for the progress of my work. I am grateful to my supervisor for the trust and autonomy given to me throughout this experience.

Finally, I would like to thank everyone who contributed, directly or indirectly, to the success of this internship and to the completion of this report.

This internship has been a valuable step in my academic and professional development, and I am grateful for the opportunity and experience it provided me.

Abstract

This report presents the work carried out during my PFA internship at **DXC Technology Morocco**. The internship was conducted remotely and was divided into two main phases. The first month was dedicated to professional training through a ServiceNow System Administrator course, which provided me with the fundamental knowledge required to understand and work with the ServiceNow platform. The second month focused on applying this knowledge through the design and implementation of a Service Request Fulfillment solution.

The project aimed to structure and automate the management of IT service requests in order to improve their processing, traceability, assignment, and overall fulfillment. Several service catalog items were designed and configured, including User Account Creation, Enterprise VPN Access, Software Installation, IT Hardware Requests, and Password Reset. The implementation involved defining request forms, approval processes, fulfillment tasks, assignment rules, notifications, and automated closure. Service Level Agreements (SLAs) were also configured to support service monitoring and ensure compliance with defined service targets.

The project was developed within the ServiceNow environment through a structured process involving requirements analysis, configuration, testing, validation, and deployment preparation. This internship provided me with practical experience in ServiceNow administration and configuration, IT Service Management, workflow automation, and requirements analysis. It also strengthened my ability to work independently, manage my time effectively, and apply newly acquired technical knowledge to a practical professional project.

Keywords: ServiceNow, IT Service Management (ITSM), Service Request Fulfillment, Service Catalog, Workflow Automation, Service Level Agreement (SLA), Business Rules

List of abbreviations

abbreviation	**Meaning**
IT	Information Technology
ITSM	Information Technology Service Management
ITIL	Information Technology Infrastructure Library
UI	User Interface
SLA	Service Level Agreement
aPaaS	Application Platform as a Service
SaaS	Software as a Service
UAT	User Acceptance Testing

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Chapter 1

General Context of the Internship

1.1 Introduction

this chapter presents the general context of my PFA Internship at **DXC Technology Morocco**. It begins with an overview of the host organization and its activities, followed by a presentation of the internship context and working environment. The chapter then introduces the main project carried out during the internship, including its background, problem statement, objectives, and scope. Finally, the methodology and general approach adopted throughout the project are presented.

1.2 Presentation of the Host Organization

1.2.1 DXC Technology Worldwide

DXC Technology is a global technology services company that helps organizations manage, modernize, and transform their information technology environments. The company provides a broad range of IT services and solutions covering applications, cloud and infrastructure, cybersecurity, analytics, workplace services, and business process services.

DXC Technology serves organizations across multiple industries and geographical regions, supporting them in addressing complex technological and operational challenges. Its activities combine consulting, technology implementation, systems integration, and managed services to support the transformation and continuous operation of enterprise IT environments.

Figure 1.1: DXC Technology Logo

1.2.1.1 Computer Sciences Corporation (CSC)

Computer Sciences Corporation (CSC) was an American information technology company specializing in technology consulting, systems integration, outsourcing, and IT services. Founded in 1959, CSC developed extensive expertise in helping organizations design, implement, and manage complex information systems.

Over the years, CSC established an international presence and provided technology services to organizations across various industries. Its experience in consulting, application services, and enterprise IT management later became an important part of the capabilities brought together within DXC Technology.

Figure 1.2: CSC logo

1.2.1.2 Hewlett Packard Enterprise (HPE)

Hewlett Packard Enterprise (HPE) is an American technology company focused on enterprise computing, infrastructure, networking, storage, and related IT services. HPE was established in 2015 following the separation of Hewlett-Packard Company into two independent organizations.

The Enterprise Services business of HPE provided organizations with services related to infrastructure management, outsourcing, applications, and enterprise technology environments. This business subsequently became one of the two major components involved in the creation of DXC Technology.

Figure 1.3: HPE logo

1.2.1.3 Creation of DXC Technology

DXC Technology was created in 2017 through the merger of Computer Sciences Corporation (CSC) and the Enterprise Services business of Hewlett Packard Enterprise (HPE). The merger combined the technological expertise, service capabilities, and international presence of both organizations to establish a new global IT services company.

DXC Technology officially began operations on April 1, 2017. Since its creation, the company has continued to develop its expertise in areas such as applications, cloud and infrastructure, cybersecurity, analytics, workplace services, and business process services.

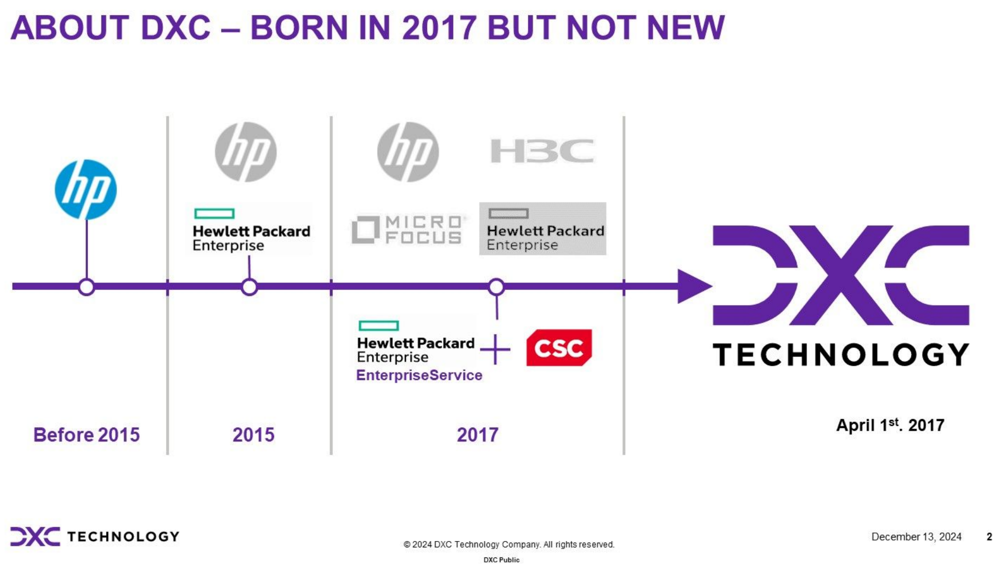

Figure 1.4: Evolution leading to the creation of DXC Technology

1.2.2 DXC Technology Morocco

DXC Technology Morocco is the Moroccan entity of DXC Technology and operates within the company's global network of technology services. The organization supports major public and private-sector organizations in their digital transformation initiatives and provides a range of technology and business services.

1.2.2.1 Overview of DXC Technology Morocco

Item	**Information**
Company	DXC Technology Morocco
Location	Technopolis, Rabat, Morocco
Established	2007
Current name	DXC Technology Morocco since 2017
Parent organization	DXC Technology
Local partner	Caisse de Dépôt et de Gestion (CDG)
Main activity	Information Technology Services

Table 1.1: DXC Technology Morocco at a Glance

1.2.2.2 History of DXC Technology Morocco

The history of DXC Technology Morocco dates back to 2007, when the Moroccan entity was established through a partnership between Electronic Data Systems (EDS) and the Caisse de Dépôt et de Gestion (CDG). The organization initially operated from a 450 m^2 site with 36 employees and began providing IT services to support CDG's activities.

In August 2008, EDS was acquired by Hewlett-Packard (HP), marking an important step in the development of the Moroccan entity. In 2009, the organization integrated HP's activities and changed its branding to HP CDG IT Services Maroc. During the same period, it obtained ISO 9001 certification.

The organization continued to expand during the following years. In 2011, it was recognized as the fifth-best employer. In 2012, the organization moved to a new site designed to accommodate up to 1,000 employees and was recognized as the third-best employer. By 2014, it had reached the second position in the best-employer ranking.

These milestones illustrate the progressive development of the Moroccan organization in terms of infrastructure, workforce, and recognition as an employer.

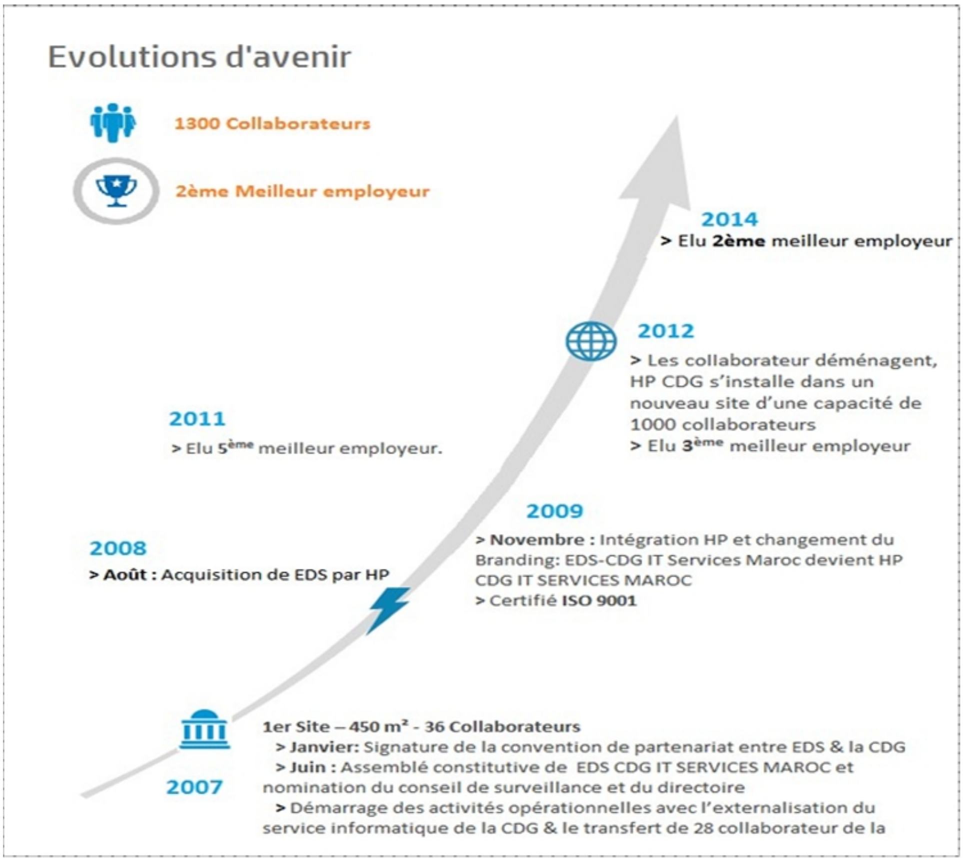

Figure 1.5: Historical development of DXC Technology Morocco

Following these developments, the organization underwent another major transformation in 2017 following the global creation of DXC Technology through the combination of Computer Sciences Corporation (CSC) and Hewlett Packard Enterprise's Enterprise Services business. The Moroccan entity subsequently adopted the DXC Technology name, becoming part of DXC Technology's global organization.

1.2.2.3 DXC Technology Success Story

The following figure presents the success story of DXC Technology and the evolution of the number of employees since 2007.

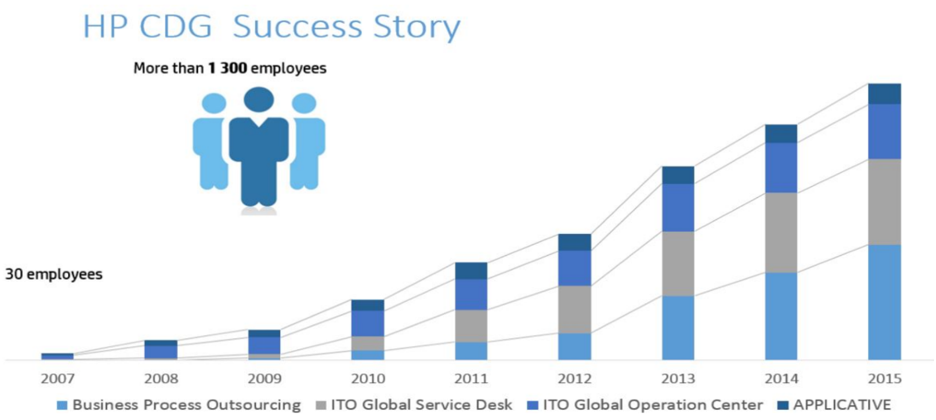

Figure 1.6: DXC Technology Success Story

1.2.2.4 Business Areas and Services

DXC Technology Morocco provides a diversified range of technical and functional services organized around several major areas of expertise. These areas combine technology capabilities with business-oriented services to address the requirements of organizations across different sectors.

- **Data Center:** Hosting, technical support, infrastructure supervision, and deployment of cloud solutions for efficient data management.
- **Cloud Infrastructure:** Design and modernization of hybrid cloud environments and migration of traditional platforms toward evolving cloud architectures.
- **Applications:** Integration, migration, modernization, and maintenance of enterprise applications.
- **ServiceNow:** Integration, customization, and optimization of business and IT processes through the ServiceNow platform, particularly in areas such as IT Service Management (ITSM) and IT Operations Management (ITOM).
- **Workplace and Mobility:** Secure access to enterprise resources through endpoint management, device management, and workplace virtualization solutions.
- **Business Process Services:** Management and support of business processes, including customer service and automated request processing.

- **Cybersecurity:** Security assessment, risk management, monitoring, incident management, and deployment of security solutions.

- **Business Intelligence and Analytics:** Development of analytical platforms and data solutions to support reporting, analysis, and decision-making.

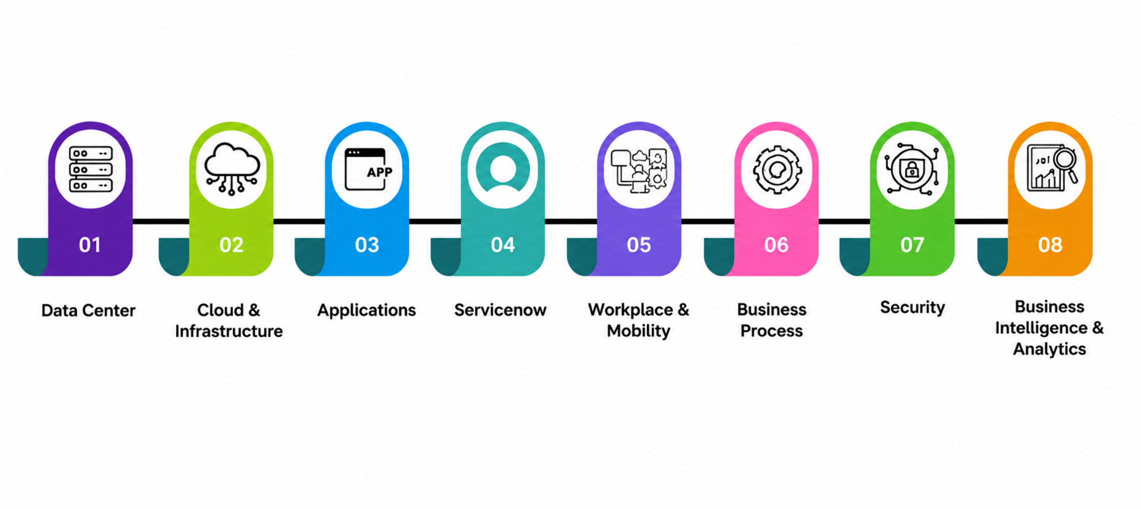

Figure 1.7: Main business areas and services of DXC Technology Morocco

1.2.2.5 Organizational Structure

DXC Technology Morocco is organized into different functional and operational structures that support the delivery of technology services to its customers. These structures include management and support functions as well as specialized service lines responsible for delivering technical and business solutions.

Figure 1.8 presents an overview of the organizational structure of DXC Technology Morocco.

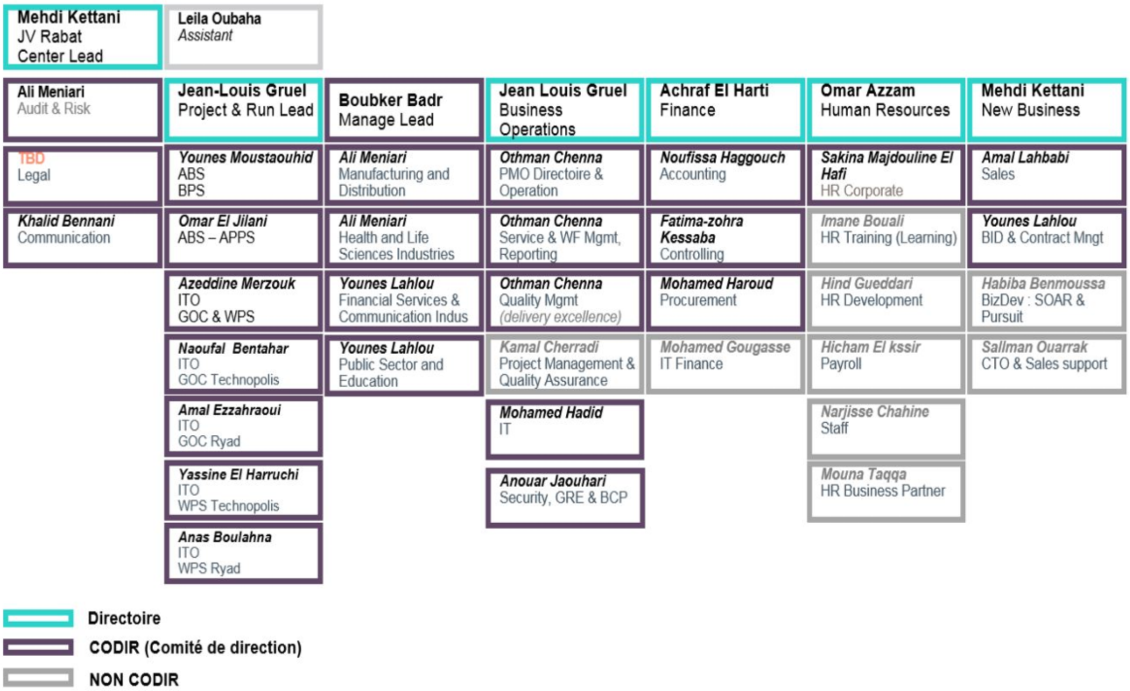

Figure 1.8: Organizational structure of DXC Technology Morocco

1.2.2.6 Clients and Strategic Partners

DXC Technology Morocco serves organizations operating across a wide range of industries, including manufacturing, retail, banking and finance, telecommunications, energy, healthcare, transportation, tourism, the public sector, and real estate.

The diversity of its customer base reflects the broad range of technology and business services provided by the organization.

Figure 1.9: Examples of DXC Technology Morocco's clients

DXC Technology Morocco also works with technology partners and maintains industry certifications that support the quality, security, and reliability of its services. These partnerships and certifications contribute to the company's ability to deliver solutions that meet technological and operational requirements.

Figure 1.10: Selected partners and certifications of DXC Technology Morocco

1.3 Internship Context

1.3.1 Internship Framework

As part of my academic curriculum, I completed a two-month PFA internship at **DXC Technology Morocco**. The internship was conducted entirely remotely and focused on the discovery and practical application of the ServiceNow platform.

The internship was supervised by **Houda AJDIR**, who provided guidance, clarifications, and feedback throughout the internship. The work was carried out independently, with regular communication with the company supervisor regarding the progress and requirements of the assigned work.

1.3.2 Internship Objectives

The main objective of this internship was to develop practical knowledge of the **ServiceNow** platform and gain experience in its administration and configuration within a professional IT environment.

More specifically, the internship aimed to:

- Acquire a solid understanding of the ServiceNow platform and its main administration concepts through the System Administrator training.
- Develop an understanding of **IT Service Management (ITSM)** concepts and their implementation within ServiceNow.
- Apply the knowledge acquired during the training to a practical ServiceNow project.

- Design and configure a structured **Service Request Fulfillment** solution for managing IT service requests.
- Gain practical experience with ServiceNow configuration components, including forms, workflows, approval processes, assignment rules, notifications, and business rules.
- Understand how automation can be used to improve the processing, traceability, and fulfillment of IT service requests.
- Develop the ability to work independently, organize tasks effectively, and manage a technical project in a remote working environment.

1.3.3 Remote Working Environment

The internship was conducted entirely remotely, without any physical presence at the company's premises. This working arrangement required me to organize my work independently and manage my time according to the objectives and requirements of the internship.

Throughout the internship, the main channel of communication was with my company supervisor, **Houda AJDIR**. She provided the necessary guidance, clarifications, and feedback regarding the training and the assigned project.

The remote nature of the internship also required a high degree of autonomy and personal organization. Tasks were carried out independently, while progress and questions were communicated to the supervisor when necessary. This environment allowed me to develop my ability to manage my workload, solve problems independently, and maintain progress throughout the different stages of the internship.

1.3.4 Internship Phases

The internship was organized into two main phases, each with a specific objective. The first phase focused on acquiring the necessary knowledge of the ServiceNow platform, while the second phase consisted of applying this knowledge to a practical project.

Phase	Duration	Main Activities
Phase 1	First Month	ServiceNow System Administrator training, covering the fundamental concepts of the platform and its administration and configuration.
Phase 2	Second Month	Design and implementation of the Service Request Fulfillment project, including the configuration and testing of the required ServiceNow components.

Table 1.2: Organization of the Internship

During the first phase, the focus was on building a solid foundation in ServiceNow administration. The training provided the knowledge required to understand the platform and its main configuration concepts.

During the second phase, the knowledge acquired during the training was applied to the development of the Service Request Fulfillment solution. This phase involved analyzing the requirements, designing the service processes, configuring the ServiceNow components, and testing the resulting solution.

1.4 Project Context

1.4.1 Background (Service Request Management)

In modern enterprise environments, Information Technology Service Management (ITSM) is critical for aligning IT services with business needs. Within the ITSM framework, Request Fulfillment is the core process responsible for managing the lifecycle of all standard service requests from users. These requests typically include access to applications, software installations, hardware provisioning, and routine information queries. An effective Request Fulfillment process ensures that employees have access to the IT resources they need to perform their daily tasks efficiently, while the organization maintains proper authorization, compliance, and security controls.

1.4.2 Problem Statement

Despite the critical nature of IT services, many organizations continue to rely on manual or fragmented processes to handle user requests. When requests are submitted via unstructured channels such as emails, phone calls, or direct messages, several operational issues arise:

- **Lack of Traceability:** It is difficult to track the status of a request, leading to poor visibility for both the end-user and the IT support team.
- **Delayed Processing:** Manual routing of requests for managerial or security approvals often creates bottlenecks, increasing the time required to fulfill the request.
- **Assignment Errors:** Without an automated routing mechanism, requests are frequently assigned to the wrong fulfillment group, causing further delays.
- **Absence of Metrics:** The inability to track request fulfillment times makes it difficult to measure performance against Service Level Agreements (SLAs) or identify areas for continuous improvement.

These challenges highlight the need for a structured and automated request fulfillment process that ensures requests are properly captured, authorized, assigned, and monitored throughout their lifecycle.

1.4.3 Project Objectives

To address these operational inefficiencies, this project aims to design and implement an automated Request Fulfillment solution using the ServiceNow platform. The primary objectives are :

- **Centralize Requests:** Provide a single, structured portal (Service Catalog) where users can easily and intuitively submit IT requests.
- **Automate Workflows:** Replace manual approval and routing steps with automated workflows to ensure requests are immediately directed to the appropriate managers and fulfillment teams.

- **Standardize Data Collection:** Use dynamic request forms to ensure all necessary information is captured at the time of submission, eliminating back-and-forth communication.
- **Enforce SLAs:** Configure Service Level Agreements to monitor fulfillment times and trigger automated escalations when targets are at risk.
- **Improve Traceability:** Provide visibility into the status and progress of requests throughout their fulfillment lifecycle.

1.4.4 Project Scope

The scope of this project encompasses the end-to-end configuration of five specific Service Catalog items that represent common and critical IT request :

1. **User Account Creation:** Automating the approval and task generation for provisioning Active Directory, email, and application accounts for new employees.
2. **Enterprise VPN Access:** Structuring the security and network approval chains required for secure remote access.
3. **Software Installation:** Managing software requests including automated license verification and installation tasks.
4. **IT Hardware Requests:** Automating the process from stock verification and procurement approval to delivery and configuration.
5. **Password Reset:** Implementing a secure, streamlined process for routine account recovery.

The technical scope involves leveraging ServiceNow's core capabilities, including the Service Catalog, Flow Designer for process automation, Business Rules and Client Scripts for backend logic, with UI Policies, User Criteria for access control, and SLA definitions.

1.4.5 Project Deliverables

The main deliverable of the project was a configured ServiceNow Request Fulfillment solution covering five IT service catalog items. The solution included structured request forms, approval and fulfillment processes, assignment mechanisms, notifications, and Service Level Agreements.

The project also resulted in configuration artifacts that could be packaged using ServiceNow Update Sets to support the controlled migration of the developed solution between environments.

1.5 Methodological Approach

To ensure the successful delivery of the Request Fulfillment solution, a structured and iterative methodological approach was adopted. The project was executed through the following key phases:

- **Requirements Analysis:** Defining the specific inputs, variables, approval chains, and fulfillment tasks required for each of the five catalog items. This phase involved translating business requirements into technical ServiceNow specifications.

- **Platform Configuration and Development:** Utilizing a ServiceNow Development instance to build the solution. This involved creating Catalog Items, defining variables and UI Policies for dynamic forms, and utilizing Flow Designer to build the underlying automation logic.

- **Testing and Validation:** Conducting rigorous testing of each workflow to ensure approvals trigger correctly, tasks are assigned to the right fulfillment groups based on the request type, and SLAs attach properly based on ticket priority.

- **Deployment Preparation:** Packaging the configurations into ServiceNow Update Sets to ensure a safe, structured, and traceable migration of the developed features from the Development environment to Testing (UAT) and Production environments.

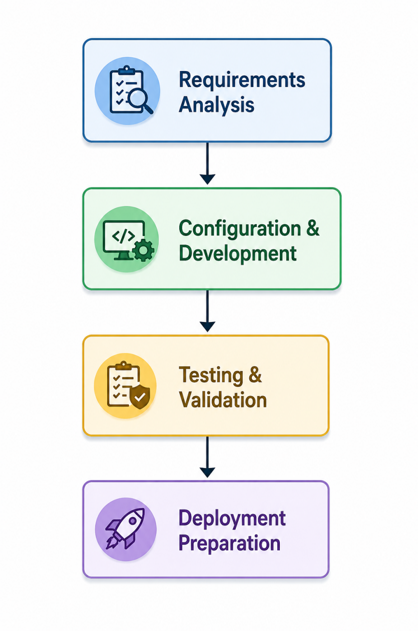

Figure 1.11: Methodological Approach Phases

1.6 Conclusion

This chapter presented the context of the internship at DXC Technology Morocco, the Service Request Fulfillment project, its objectives and scope, and the methodological approach adopted. The following chapter introduces the ServiceNow platform, IT Service Management (ITSM), and the main technical concepts used in the project.

Chapter 2

Theoretical and Technical Background

2.1 Introduction

This chapter presents the theoretical and technical concepts required to understand the Service Request Fulfillment solution developed during the internship. It introduces the ServiceNow platform, its architecture, and its core capabilities. Furthermore, it explores the principles of Information Technology Service Management (ITSM) and the role of ITIL best practices in service request management. Finally, the chapter details the main ServiceNow components used to design, automate, and prepare the deployment of the solution, including the Service Catalog, Flow Designer, Service Level Agreements (SLAs), and the ServiceNow development lifecycle.

2.2 ServiceNow Platform

Figure 2.1: ServiceNow Logo

2.2.1 Overview of ServiceNow

ServiceNow is a cloud-based enterprise platform that provides Software as a Service (SaaS) and Application Platform as a Service (aPaaS) solutions. It is designed to support the management and automation of enterprise processes and services. Initially focused on Information Technology Service Management (ITSM), the platform has progressively expanded its capabilities to cover different areas of an organization, including IT operations, security, human resources, customer service, and business processes.

The platform provides a centralized environment in which organizations can define processes, manage requests, automate tasks, and monitor their execution. Its configurable architecture also allows organizations to adapt the platform to their specific business and operational requirements.

Founded	2004
Founded by	Fred Luddy
Headquarters	Santa Clara, California, USA
Logo	servicenow
Website	Servicenow.com

Table 2.1: ServiceNow Overview

2.2.2 ServiceNow Architecture

ServiceNow is based on a cloud architecture designed to provide each customer with and isolated and configurable environment. One of the main characteristics of the platform is its **multi-instance architecture**, in which each customer is provided with a dedicated instance.

Unlike a traditional multi-tenant architecture, where multiple customers share the same application environment, the multi-instance approach provides a greater level of isolation between customer environments. Each instance contains its own application, middleware, and database resources, while the underlying physical infrastructure can be shared.

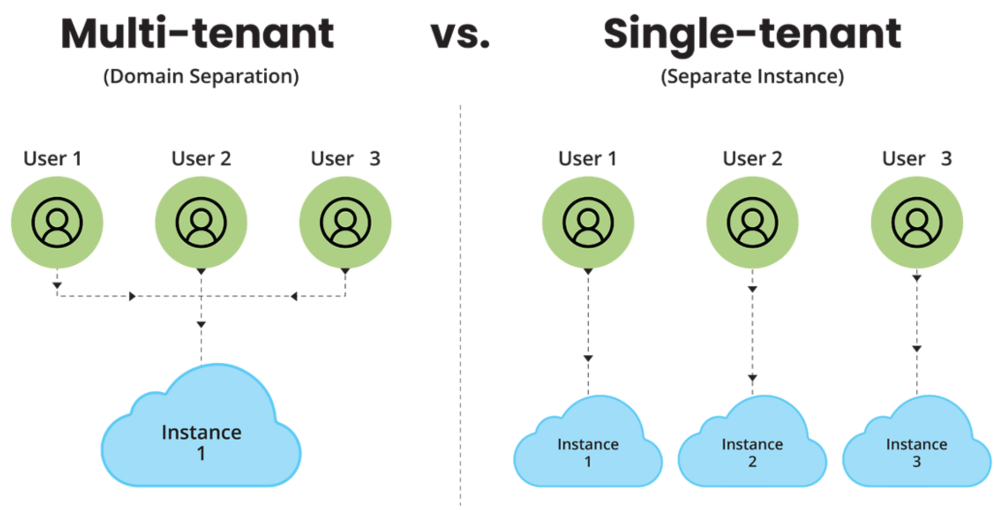

Figure 2.2: Multi Tenant vs Single Tenant

This architecture provides several advantages, particularly in terms of data isolation, configuration flexibility, and platform management. changes or configurations made within one instance are isolated from other customer instances, allowing organizations to adapt the platform to their specific requirements.

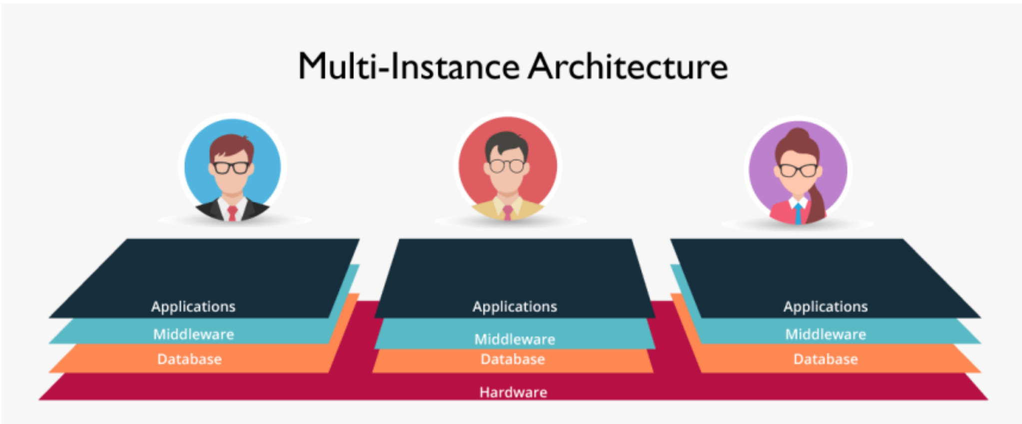

Figure 2.3: Multi-Instance Architecture of ServiceNow

2.2.3 ServiceNow Applications and Modules

ServiceNow provides a wide range of applications and modules designed to support different organizational processes and address specific enterprise needs :

- **ITSM (IT Service Management):** The core module focused on delivering and managing IT services, encompassing incident, problem, change, and request management.

- **ITOM (IT Operations Management):** Focuses on infrastructure visibility, service health, and cloud provisioning.
- **ITBM (IT Business Management):** Aligns IT investments with business goals through project portfolio management and financial tracking.
- **Security Operations:** Integrates security incident response and vulnerability management directly into IT workflow.

This project is built primarily on the ITSM module, leveraging its Service Catalog and workflow automation capabilities.

2.2.4 ServiceNow Development and Configuration

ServiceNow is a low-code/no-code platform that allows organizations to build and customize applications rapidly. Many processes can be configured using graphical interfaces and predefined platform components, reducing the need for extensive programming. However, the platform also provides scripting capabilities for implementing more complex business logic and extending standard functionality.

JavaScript is the primary scripting language used within ServiceNow. It can be used to implement server-side logic, control client-side behavior, validate data, and automate specific operations according to business requirements. This combination of configuration and scripting allows developers to adapt the platform while maintaining a flexible and scalable development environment.

For the Service Request Fulfillment project, ServiceNow configuration was mainly used to create and customize Service Catalog items, define request variables, control form behavior, configure approval and fulfillment processes, establish assignment rules and notifications, and implement Service Level Agreements (SLAs). These elements provided the technical foundation for the automation of the request fulfillment processes.

2.3 Information Technology Service Management (ITSM)

2.3.1 ITSM Overview

Information Technology Service Management (ITSM) is a strategic approach to designing, delivering, managing, and improving the way information technology services are provided within an organization. The primary objective of ITSM is to ensure that the appropriate processes, people, and technologies are in place to support the organization in achieving its business goals. This approach shifts the focus from managing isolated IT infrastructure components to delivering integrated and effective IT services.

2.3.2 ITIL and Service Management

ITIL(Information Technology Infrastructure Library) is a widely adopted framework that provides guidance and best practices fir IT Service Management. It helps organizations establish structured approaches for designing, delivering, supporting, and continually improving IT-enabled services.

ITIL emphasizes the importance of defining clear processes, responsibilities, and service objectives. Its practices can be adapted to the specific requirements of an organization rather than being implemented as a rigid set of procedures.

For this project, ITIL concepts provide a general framework for understanding service request management and the importance of standardized processes, defined approval procedures, controlled fulfillment, and service-level monitoring.

2.3.3 Service Request Management

Service Request Management is an ITSM practice focused on handling standard requests submitted by users for predefined IT services. These requests may include application access, software installation, hardware requests, or password resets.

A structured request management process ensures that requests are properly recorded, validated, authorized, assigned, fulfilled, and closed. This improves the consistency and traceability of service delivery while ensuring that requests follow predefined procedures.

In ServiceNow, Service Request Management is supported by the **Service Catalog**, which provides users with a structured interface for submitting requests. Each catalog item defines the required information and can be associated with approval processes, fulfillment tasks, assignment rules, notifications, and automation.

The process can therefore be summarized through several main activities: request submission, approval, assignment, fulfillment, and closure. In the context of this project, five types of IT service requests were considered:

1. **User Account Creation**
2. **Enterprise VPN Access**
3. **Software Installation**
4. **IT Hardware Requests**
5. **Password Reset**

These request types form the basis of the Service Request Fulfillment solution developed during the internship.

2.3.4 Request Fulfillment Lifecycle

The fulfillment of an IT service request follows a structured lifecycle that ensures the request is properly processed from submission to completion. A typical request fulfillment process consists of several stages, including submission, approval, assignment, fulfillment, and closure.

1. **Request Submission:** The user selects the required service and provides the necessary information through the Service Catalog.
2. **Approval:** The request is reviewed and, when required, submitted to the appropriate person or department for approval.
3. **Assignment:** Once approved, the request is assigned to the appropriate fulfillment group or generates the required fulfillment tasks.
4. **Fulfillment:** The responsible team performs the activities required to provide the requested service.

5. **Closure:** After the required activities are completed, the request is validated and closed.

In ServiceNow, these stages can be supported and automated through workflows, approvals, task assignment, notifications, and other platform capabilities. This automation helps reduce manual processing, improve traceability, and ensure that requests follow a consistent fulfillment process.

Figure 2.4 illustrates the main stages of the request fulfillment lifecycle.

Figure 2.4: Request Fulfillment Lifecycle in ServiceNow

2.4 ServiceNow Service Catalog

2.4.1 Service Catalog and Catalog Items

The Service Catalog is the user-facing interface of the ITSM module, providing a centralized portal where employees can browse, search for, and request predefined IT services. It standardizes the submission of service requests by providing structured forms and ensuring that the necessary information is collected from the beginning of the request process.

Catalog Items represent the individual services available for request within the Service Catalog. Each item is designed to address a specific user need and can be organized into appropriate categories to facilitate navigation and selection.

For this project, five Catalog Items were designed to address common IT service needs: **User Account Creation**, **Enterprise VPN Access**, **Software Installation**, **IT Hardware Requests**, and **Password Reset**. Each item was configured with the information and process requirements necessary for its subsequent fulfillment.

2.4.2 Catalog Configuration and Access Control

ServiceNow provides several configuration features that allow Catalog Items to be adapted to the specific requirements of each service. One of the main elements is the use of **variables**, which define the information that users must provide when submitting a request. Variables can be used to collect different types of information, such as text, selections, dates, or references to existing records.

Variable Sets can also be used to group related variables and reuse them across multiple Catalog Items. This helps maintain consistency and reduces the need to configure the same information repeatedly.

The behavior and visibility of form fields can be controlled using **UI Policies**. These policies allow fields to be made mandatory, visible, or read-only depending on the conditions defined for a request. This makes it possible to create dynamic forms that display only the information relevant to the selected service.

Access to Catalog Items can be managed using **User Criteria**, which define which users or groups are authorized to access specific services. This ensures that catalog services are presented only to the appropriate users according to the organization's access requirements.

For the Service Request Fulfillment project, these configuration mechanisms were used to structure the request forms, collect the required information, control form behavior, and manage access to the different Catalog Items.

2.4.3 Catalog Configuration and Access Control

To collect the information required for a specific request, Catalog Items use **Variables** and **Variable Sets**, which act as input fields on the request form. These elements allow each Catalog Item to collect the information needed for its specific fulfillment process.

To provide a consistent user experience, **UI Policies** and **Client Scripts** can be used to dynamically control form fields, making them visible, mandatory, or read-only according to predefined conditions. In addition, **User Criteria** can be used to control access to Catalog Items and ensure that services are available only to authorized users.

For the Service Request Fulfillment project, these configuration features were used to structure the request forms, collect the required information, control form behavior, and manage access to the different services.

2.5 Workflow Automation

2.5.1 Flow Designer and Process Automation

Flow Designer is ServiceNow's visual automation tool for creating multi-step processes. It allows developers to define a sequence of actions that are automatically executed when a predefined trigger occurs. These actions can include creating tasks, requesting approvals, updating records, and sending notifications.

Flow Designer helps replace manual routing and processing by connecting the different stages of a service request into a structured workflow. In the Service Request Fulfillment project, it was used to automate the processing of requests and coordinate their approval and fulfillment activities.

ServiceNow also supports backend automation through **Business Rules**. These server-side scripts execute when records are inserted, updated, or queried, allowing additional business logic to be applied to the system. Together, Flow Designer and Business Rules provide mechanisms for automating both process-level and record-level operations.

2.5.2 Approvals, Tasks, and Notifications

Automated workflows rely on approvals, fulfillment tasks, and notifications to coordinate the different stages of a request.

- **Approvals:** Approval requests are generated when a service requires authorization before fulfillment can continue. The approval process can involve managers or other responsible teams depending on the type of request.

- **Catalog Tasks (SCTASKs):** Once the required approvals are completed, fulfillment tasks can be generated and assigned to the appropriate support groups. These tasks represent the activities required to complete the requested service.

- **Notifications:** Automated notifications keep users and stakeholders informed about important events, such as pending approvals, task assignments, status changes, and request completion.

These mechanisms work together to provide a structured and traceable fulfillment process.

2.6 Service Level Agreements

2.6.1 SLA Concepts

A **Service Level Agreement (SLA)** is a documented commitment that defines the expected level of service and measurable performance targets. SLAs commonly specify response or fulfillment times and provide a basis for monitoring service performance.

In IT Service Management, SLAs help establish clear expectations, measure performance, and identify requests that are approaching or exceeding their defined service targets.

2.6.2 SLA Management in ServiceNow

In ServiceNow, **SLA Definitions** establish the conditions used to track the time associated with a record. These definitions can specify when an SLA starts, pauses, resumes, and stops according to predefined conditions.

ServiceNow continuously tracks active SLAs and provides visibility into the remaining time and current status of each target. This allows support teams to monitor fulfillment performance and identify requests that may require attention before an SLA is breached.

For the Service Request Fulfillment project, SLAs were considered to monitor the time required to process service requests and to support compliance with defined fulfillment targets.

2.7 Development and Deployment

2.7.1 Development and Testing Environments

ServiceNow development commonly uses separate environments for development, testing, and production. This separation allows configurations to be developed and validated before they are made available to end users.

The main environments used in a typical ServiceNow development lifecycle are:

- **DEV:** The development environment where configurations, Catalog Items, forms, and workflows are created and modified.
- **TEST (UAT):** The testing environment used to validate the developed solution and verify that it meets the defined requirements.
- **PROD:** The production environment where the validated solution is made available to end users.

This separation contributes to a controlled development process and reduces the risk of introducing untested changes into the production environment.

2.7.2 Update Sets and Deployment

An **Update Set** is a ServiceNow mechanism used to capture and organize configuration changes made within an instance. It allows related customizations to be packaged and transferred between ServiceNow environments.

Update Sets can contain different configuration elements, such as Catalog Items, workflows, business rules, client scripts, and UI policies. They therefore facilitate the controlled migration of developed configurations from a development environment to a testing or production environment.

For this project, Update Sets were used to organize the developed configurations and prepare them for controlled migration between the different ServiceNow environments.

2.8 Conclusion

This chapter presented the main theoretical and technical concepts underlying the Service Request Fulfillment project. It introduced ServiceNow, ITSM, Service Request Management, the Service Catalog, workflow automation, SLAs, and the development and deployment process. These concepts provide the foundation for the following chapters, which focus on the design and practical implementation of the developed solution.

Chapter 3

Analysis and Design of the Service Request Fulfillment Solution